

COSC 594

DM Composition (I)

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Outline

- MOTIVATION
- EXAMPLES
- DIFFUSION MODEL COMPOSITION

Outline

■ MOTIVATION

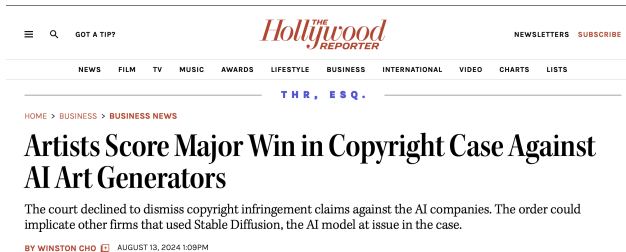
The cost of training DMs

■ COMPUTATIONAL COST

- ★ Stable Diffusion: $\approx 150,000$ A100 GPU hours
- ★ DALL-E: $150,000 \sim 200,000$ A100 GPU hours
- ★ Open-Sora: $\approx 28,000$ H100 GPU hours

data source: Wikipedia and guess

■ DATA COST



The screenshot shows the top portion of a news article on The Hollywood Reporter's website. The header includes the site's logo, navigation links for various entertainment categories, and a search bar. The article title is prominently displayed in a large, bold serif font. Below the title, a short introductory paragraph is visible, followed by the byline 'BY WINSTON CHO' and the date 'AUGUST 13, 2024 1:09PM'.

THE HOLLYWOOD REPORTER

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THR, ESQ.

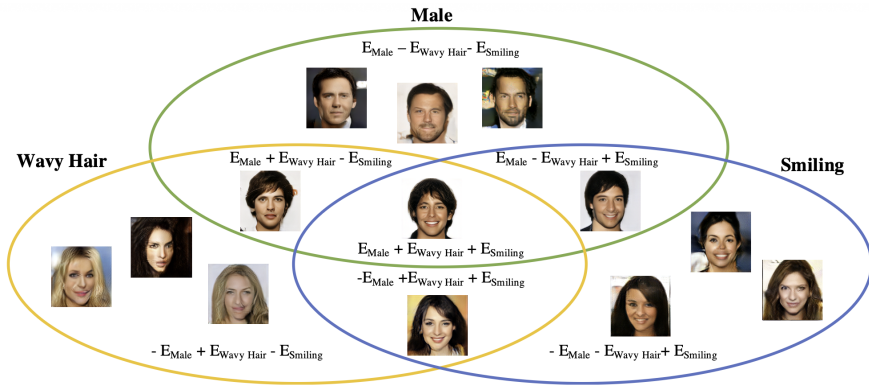
HOME > BUSINESS > BUSINESS NEWS

Artists Score Major Win in Copyright Case Against AI Art Generators

The court declined to dismiss copyright infringement claims against the AI companies. The order could implicate other firms that used Stable Diffusion, the AI model at issue in the case.

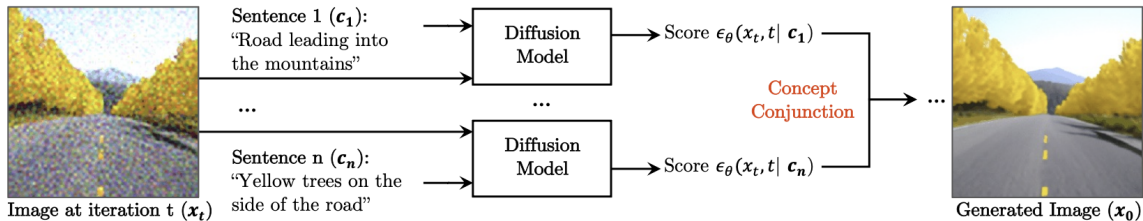
BY WINSTON CHO AUGUST 13, 2024 1:09PM

Generalization



Concept composition

Unseen environment

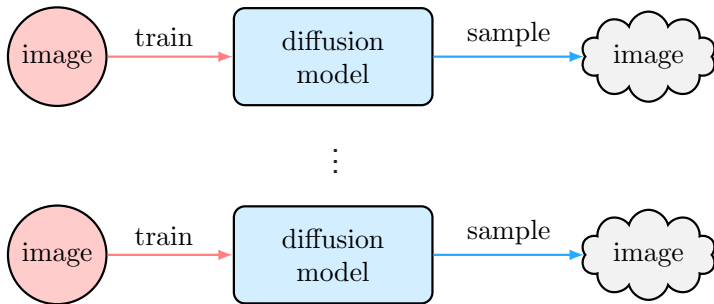


Compositional generation

Combination of DMs



⋮



$$x_0 \sim p^1(\cdot), \dots, p^K(\cdot)$$

How do we combine several pretrained DMs?

Outline

■ MOTIVATION

■ EXAMPLES

Energy-based model

Energy function: $E_{\theta}(x): \mathbb{R}^d \rightarrow \mathbb{R}$ w/ parameter θ

Energy-based model: $p_{\theta}(x) \propto e^{-E_{\theta}(x)}$

■ LANGEVIN DYNAMICS

$$x_t = x_{t-1} - \eta \underbrace{\nabla_x E_{\theta}(x_{t-1})}_{\text{score-like function}} + \sqrt{2\eta} \epsilon_t \quad \text{where } \epsilon_t \sim \mathcal{N}(0, I)$$

As $t \rightarrow \infty$, dynamics converge to samples $x_t \sim p_{\theta}(\cdot)$

Energy model composition: conjunction

K energy-based models: $p^k(x) \propto e^{-E^k(x)}$ for $k = 1, \dots, K$

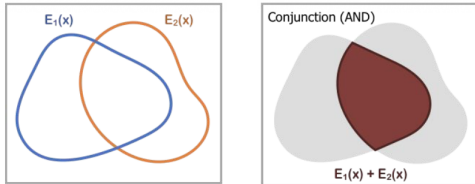


image source: arXiv:2004.06030

■ COMPOSED MODEL

$$p^{\text{compose}}(x) \propto p^1(x)p^2(x)\dots p^K(x) \propto e^{-\sum_{k=1}^K E^k(x)}$$

$$\Rightarrow x_t = x_{t-1} - \eta \sum_{k=1}^K \nabla_x E^k(x_{t-1}) + \sqrt{2\eta} \epsilon_t \quad \text{where } \epsilon_t \sim \mathcal{N}(0, I)$$

Energy model composition: disjunction

K energy-based models: $p^k(x) \propto e^{-E^k(x)}$ for $k = 1, \dots, K$

normalization constants are approximately equal

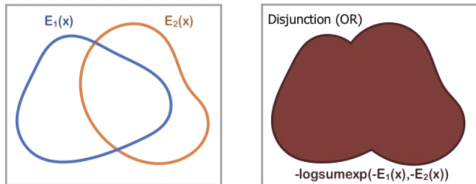


image source: arXiv:2004.06030

■ COMPOSED MODEL

$$p^{\text{compose}}(x) \propto \sum_{k=1}^K p^k(x) \propto e^{\log \sum_{k=1}^K e^{-E^k(x)}}$$

$$\implies x_t = x_{t-1} + \eta \nabla_x \log \sum_{k=1}^K e^{-E^k(x)} + \sqrt{2\eta} \epsilon_t \quad \text{where } \epsilon_t \sim \mathcal{N}(0, I)$$

Energy model composition: negation

K energy-based models: $p^k(x) \propto e^{-E^k(x)}$ for $k = 1, \dots, K$

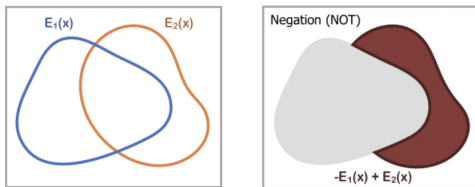


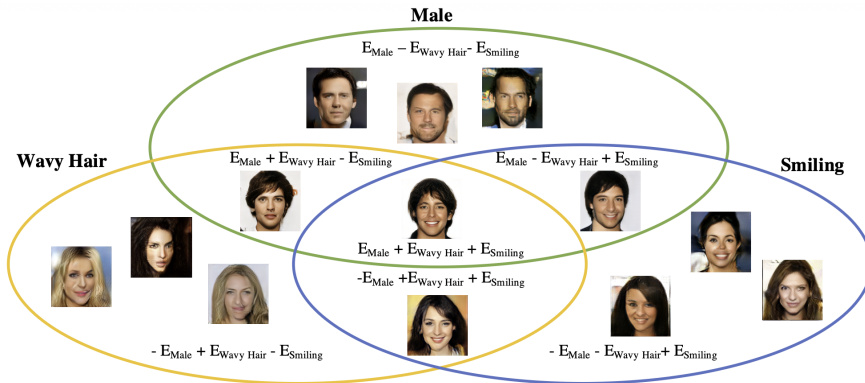
image source: arXiv:2004.06030

■ COMPOSED MODEL

$$p^{\text{compose}}(x) \propto \frac{p^1(x)}{p^2(x)} \propto e^{-E^1(x) + E^2(x)}$$

$$\Rightarrow x_t = x_{t-1} - \eta \nabla_x (E^1(x_{t-1}) - E^2(x_{t-1})) + \sqrt{2\eta} \epsilon_t \quad \text{where } \epsilon_t \sim \mathcal{N}(0, I)$$

Generalization of CelebA dataset

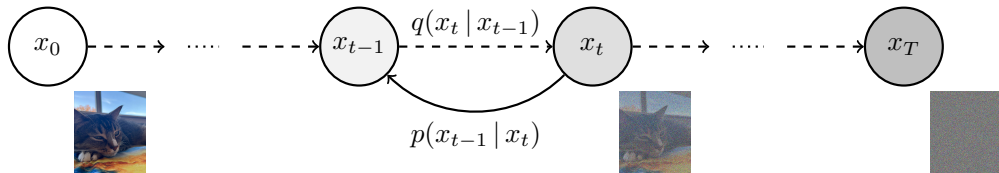


Concept conjunction, disjunction, and negation

Outline

- MOTIVATION
- EXAMPLES
- DIFFUSION MODEL COMPOSITION

Diffusion model



data point x_0 – image, trajectory, video, protein

■ ANALOGUE W/ VAE

- ★ input and latent

$$x = x_0 \quad \text{and} \quad z_{1:T} = x_{1:T}$$

- ★ noiser (encoder)

$$q(x_{1:T} | x_0) = q(x_1 | x_0)q(x_2 | x_1) \dots q(x_T | x_{T-1})$$

- ★ denoiser (decoder)

$$p(x_0 | x_{1:T}) = p(x_T)p(x_{T-1} | x_T) \dots p(x_0 | x_1)$$

■ GROUND-TRUTH DENOISER

$$q(x_{t-1} | x_t, x_0) = \mathcal{N}(x_{t-1}; \mu_q(x_t, x_0), \sigma^2(t)I)$$

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{\alpha_t}x_{t-1}, (1 - \alpha_t)I)$$

$$\mu_q(x_t, x_0) = \frac{1}{\sqrt{\alpha_t}}x_t + \frac{1-\alpha_t}{\sqrt{\alpha_t}}\nabla \log p(x_t) - \text{score-based mean}$$

$$\sigma_q^2(t) = \frac{(1-\alpha_t)(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} - \text{variance}$$

■ APPROXIMATE DENOISER

$$p_\theta(x_{t-1} | x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \sigma^2(t)I) \quad \text{where} \quad \mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}}x_t + \frac{1-\alpha_t}{\sqrt{\alpha_t}}s_\theta(x_t, t)$$

■ MEAN SQUARE MINIMIZATION

$$\underset{\theta}{\text{minimize}} \quad D_{\text{KL}}(q(x_{t-1} | x_t, x_0) \| p_\theta(x_{t-1} | x_t)) \quad \Leftrightarrow \quad \underset{\theta}{\text{minimize}} \quad \|\nabla \log p(x_t) - s_\theta(x_t, t)\|^2$$

$$\text{Pretrained model } p_t^{\text{pre}}(x_{t-1} | x_t) = p_{\theta^*}(x_{t-1} | x_t)$$

Diffusion model as energy-based model

Generative process of diffusion model: $x_{t-1} = \frac{1}{\sqrt{\alpha_t}}x_t + \frac{1-\alpha_t}{\sqrt{\alpha_t}}s_{\theta^*}(x_t, t) + \sigma(t)\epsilon_t$

Optimal score function: $s_{\theta^*}(x_t, t) = \nabla \log p(x_t)$

■ ENERGY-BASED DISTRIBUTION

$$p(x) \propto e^{-E(x)} \implies \nabla \log p(x) = -\nabla_x E(x)$$

★ Langevin dynamics

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}}x_t - \frac{1-\alpha_t}{\sqrt{\alpha_t}}\nabla_x E(x_t) + \sigma(t)\epsilon_t$$

Score function as negative gradient of energy function

Diffusion model composition: conjunction

K diffusion models: $x_{t-1} = \frac{1}{\sqrt{\alpha_t}}x_t + \frac{1-\alpha_t}{\sqrt{\alpha_t}}\nabla \log p^k(x_t) + \sigma(t)\epsilon_t$ for $k = 1, \dots, K$

■ DISTRIBUTION CONJUNCTION

$$p^{\text{compose}}(x) \propto \prod_{k=1}^K p^k(x) \implies \nabla \log p^{\text{compose}}(x) = \sum_{k=1}^K \nabla \log p^k(x)$$

★ composed generative process

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}}x_t + \frac{1-\alpha_t}{\sqrt{\alpha_t}} \sum_{k=1}^K \nabla \log p^k(x_t) + \sigma(t)\epsilon_t$$

Conjunction via **adding up score functions**

Diffusion model composition: negation

K diffusion models: $x_{t-1} = \frac{1}{\sqrt{\alpha_t}}x_t + \frac{1-\alpha_t}{\sqrt{\alpha_t}}\nabla \log p^k(x_t) + \sigma(t)\epsilon_t$ for $k = 1, \dots, K$

■ DISTRIBUTION NEGATION

$$p^{\text{compose}}(x) \propto \frac{p^1(x)}{p^2(x)} \implies \nabla \log p^{\text{compose}}(x) = \nabla \log p^1(x) - \nabla \log p^2(x)$$

★ composed generative process

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}}x_t + \frac{1-\alpha_t}{\sqrt{\alpha_t}}(\nabla \log p^1(x_t) - \nabla \log p^2(x_t)) + \sigma(t)\epsilon_t$$

Negation via **subtracting a score function**

Controlled generation

(a) Composing Language Descriptions (Composed Stable Diffusion)



"A photo of cherry blossom trees" AND "Sun dog" AND "Green grass"



"A church" AND "Lightning in the background" AND "A beautiful pink sky"



"A stone castle surrounded by lakes and trees," AND "Black and white"



"A stone castle surrounded by lakes and trees," AND (NOT "Black and white")



"A mystical tree " AND "A dark magical pond" AND "Dark"



"A mystical tree " AND "A dark magical pond" AND (NOT "Dark")

(b) Composing Language Descriptions (Composed GLIDE)



"A red car parked in a desert" AND "hills behind the car" AND "Aurora in the sky"



"The sun setting in a horizon" AND "A house next to a pond" AND "Hills in the background"



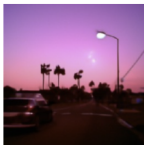
"A house with snow on the roof" AND "The house behind a tree" AND "A car in front of a tree"



"A cloudy blue sky" AND "A mountain in the horizon" AND "Cherry Blossoms in front of the mountain"



"A Ferris wheel" AND "A lake right next to the Ferris wheel" AND "Buildings next to the lake"



"Palm trees on both sides of the street" AND "Pink Sunset in the horizon" AND "A car moving away"

Diffusion model composition: disjunction

K diffusion models: $x_{t-1} = \frac{1}{\sqrt{\alpha_t}}x_t + \frac{1-\alpha_t}{\sqrt{\alpha_t}}\nabla \log p^k(x_t) + \sigma(t)\epsilon_t$ for $k = 1, \dots, K$

■ DISTRIBUTION CONJUNCTION

$$p^{\text{compose}}(x) \propto \sum_{k=1}^K p^k(x) \implies \nabla \log p^{\text{compose}}(x) \neq \sum_{k=1}^K \nabla \log p^k(x)$$

★ normalization constant matters

$$\nabla \log p^{\text{compose}}(x) = \nabla \log \sum_{k=1}^K \frac{1}{Z^k} p^k(x)$$

Disjunction needs **mixture weights**

Summary

■ MOTIVATION

Training cost, generalization

■ EXAMPLES

Energy-based model composition: conjunction, negation, disjunction

■ DIFFUSION MODEL COMPOSITION

Arithmetic operations for conjunction and negation, but disjunction